NET REVENUE MANAGEMENT IN EXCEL

Case Study Report | FMCG — Shampoo Category | HealthMax

Executive Summary

This report documents a comprehensive Net Revenue Management (NRM) analysis conducted for HealthMax, a market-leading FMCG company in the shampoo industry. The project was structured across three chapters covering market analysis, NRM opportunity identification, and strategic recommendations. Working with multi-year transaction data spanning 2018–2023, the analysis combined KPI modeling, pricing strategy, mix management, and forecasting to identify concrete revenue growth initiatives worth an estimated $602,857 in incremental 2024 net sales above the organic growth baseline.

Project Overview

Attribute	Details
Company	HealthMax (fictional FMCG, shampoo market leader)
Industry	Fast-Moving Consumer Goods (FMCG)
Dataset	Unit & value sales by Brand, Region, Month, Year (2018–2023)
Tool	Microsoft Excel (PivotTables, SUMIFS, VLOOKUP, Forecast Sheet, Charts)
Scope	Full category analysis + internal portfolio review + strategic NRM initiatives
Output	Market KPIs, profitability matrix, product launch model, promotion ROI, 2024 forecast

Chapter 1: Market Analysis & KPI Framework

The first phase established a rigorous analytical foundation by calculating key performance indicators across multiple time dimensions using the full category dataset.

1.1 Dataset Sense-Check & Preparation

Before any analysis, the raw dataset was audited for data quality issues. The Values Month column contained hardcoded dollar signs ($) that prevented numeric calculations. These were stripped and the column was reformatted with proper currency formatting. A PivotTable was then created to inventory all brands by supplier, confirming HealthMax owns 2 brands in the market: Starbust and Shinez.

1.2 Year-Over-Year Growth

A PivotTable filtered to HealthMax brands was created to display value sales per brand across all available years. The Show Values As > % Difference From (Previous Year) feature was applied to surface YoY growth rates. The 2023 data was excluded from comparison since only 3 months were available, making a full-year comparison misleading.

Key finding:

- HealthMax's strongest growth year was 2020, with a category-level increase of 4%
- Shinez drove this outperformance with 7.7% YoY growth in 2020

1.3 Year-to-Date (YTD) Sales

A new YTD column was added to the dataset using a multi-criteria SUMIFS formula that aggregates unit and value sales for matching Brand, Region, and Year, filtered to months on or before the current row's month:

```
=SUMIFS(Values_Month, Brand, [@Brand], Region, [@Region], Year, [@Year], Month,
"<="&[@Month])
```

Assumption: YTD is calculated at the Brand × Region × Year level of granularity, consistent with the raw data structure. Results were manually verified by cross-checking summed monthly values against YTD figures for selected rows.

Validation output: YTD value sales for RedRose, Central region, July 2018 = $452,940.

1.4 Moving Annual Total (MAT)

MAT represents the rolling sum of the prior 12 months. It was calculated by combining the YTD values for the current year with a second SUMIFS that captures the remaining months from the prior year:

```
=[@[Units YTD]] + SUMIFS(Units_Month, Brand, [@Brand], Region, [@Region], Year,
[@Year]-1, Month, ">"&[@Month])
```

This formula logic ensures that for any given row, the MAT captures exactly 12 months of data regardless of the current month. The latest available MAT (March 2023) was used to size the total shampoo category.

Key finding: Total shampoo category turnover in the latest 12 months = $98,452,906.

1.5 Market Share Analysis

A PivotTable was built with Year in Rows and Brand in Columns, displaying value sales as % of Row Total to derive annual market share by brand. Brands were sorted by size (largest to smallest) and a line chart was added to visualize evolution over time. A Region slicer was connected to the PivotTable to enable dynamic filtering.

Key finding:

- Starbust (HealthMax) holds its strongest market share in the Central region, reaching 21.59% in 2023

1.6 Strategic Interpretation: New Strategy Assessment

A business scenario was evaluated: FreshCo Industries shifted all spend from TV to digital advertising in January 2023. By March 2023, their monthly market share was 26%, YTD was

28%, and MAT was 30%. The declining trend from MAT (30%) to YTD (28%) to monthly (26%) signals that their market share is deteriorating, not growing. Recommendation: do not replicate FreshCo's strategy.

Chapter 2: NRM Opportunity Identification

Chapter 2 introduced the core Net Revenue Management framework and applied two of its five pillars — Brand Portfolio Pricing and Mix Management — to HealthMax's internal sales data.

2.1 Net Revenue Management: Framework Overview

NRM is a structured approach to growing profitable revenue by optimizing price, mix, packaging, promotions, and trade terms across a portfolio. The five pillars applied in this project are:

- Brand Portfolio Pricing — setting the right price relative to consumer willingness-to-pay
- Mix Management — optimizing portfolio composition toward higher-margin products
- Pack Price Architecture — designing size/price ladders to capture new consumer segments
- Trade Terms Management — allocating investment across customers and channels
- Promotion Management — measuring and optimizing promotional ROI

2.2 Net Sales Calculation

Net sales were calculated for each product in HealthMax's 2022 portfolio using the formula:

```
Net Sales 2022 = Net Price × Volume 2022
```

Note: Net Price reflects the price HealthMax receives after trade discounts (i.e., it is not the retail price). Total HealthMax Net Sales 2022 = $20,480,714.90.

2.3 Gross Margin Analysis

Two additional profitability metrics were computed per product:

Metric	Formula
Gross Profit per Unit	Net Price − COGS
Gross Profit per Product	Gross Profit per Unit × Volume 2022
Gross Margin	Gross Profit per Unit ÷ Net Price
Weighted Avg. Gross Margin	Total Gross Profit ÷ Total Net Sales

Key finding: HealthMax's weighted average gross margin across the shampoo portfolio = 68.49%. This indicates strong underlying unit economics — approximately $0.69 of every dollar of net revenue flows through to gross profit.

2.4 Profitability Matrix

A Scatter Plot was built to visualize the Net Sales Contribution vs. Gross Margin trade-off across all products. Since Excel does not support direct PivotTable-to-scatter-plot conversion, the PivotTable data was copied and pasted as values before charting. Axes were configured with

Gross Margin on the horizontal axis and Net Sales Contribution on the vertical axis, producing a 2×2 priority matrix for portfolio investment decisions.

Key finding: The Starbust Strong Hair range sits in the bottom-left quadrant — low margin, low contribution — making it a lower investment priority relative to other products in the portfolio.

2.5 Mix Management: New Category Opportunity

HealthMax participates in only 2 of the 5 shampoo subcategories. A PivotTable was constructed to measure unit growth by subcategory from 2018 to 2022 using the Show Values As > % Difference From 2018 method.

Key finding: The Organic subcategory grew by 275.88% in units between 2018 and 2022 — the fastest-growing segment by a significant margin.

To size the 2024 opportunity, the Units MAT for Organic shampoos in March 2023 was identified, then projected forward:

```
Estimated 2024 Units = Units MAT (March 2023) × (1 + 20%)
```

Assumption: A 20% growth rate was applied based on observed historical trajectory as communicated by category management.

Estimated 2024 Organic shampoo units sold = 1,020,877.

2.6 New Product Launch Evaluation

Two candidate products for the 2024 organic launch were evaluated: Herbashine and HerbEssentials. Estimated financials were modeled for both using the 2024 category size estimate and each product's assumed market share, net price, and COGS:

Metric	Herbashine	HerbEssentials
Est. Market Share	Higher Margin	Higher Volume
Net Sales (Est.)	Lower	~$0.34M+
Gross Profit	Higher	Lower
Gross Margin	Superior	Inferior

Recommendation: Launch Herbashine. Despite generating lower net sales than HerbEssentials, it delivers greater gross profit — the primary decision criterion for HealthMax management.

Chapter 3: Revenue Optimization & Forecasting

Chapter 3 completed the NRM framework by addressing the remaining three pillars and consolidating all initiatives into a 2024 revenue forecast presented as a waterfall chart.

3.1 Pack Price Architecture: 50ml Shampoo

Research indicated unmet demand for travel-sized (50ml) shampoo — a format absent from HealthMax's existing portfolio. The pricing strategy for the new SKU was derived from the per-ml price of the best-rotating 100ml product (Starbust Ultra Soft), plus a 50% size premium:

```
Retail Price (50ml) = 50ml × (Retail Price (100ml) / 100ml) × (1 + 50%)
```

Assumption: Volume for the 50ml SKU was estimated at 10% of the 100ml product's 2022 volume, reflecting a conservative market entry estimate. Net Price = $2.30; COGS = $0.70.

Key finding: The 50ml format would generate $265,960 in incremental net sales.

3.2 Promotion Management: ROI Analysis

HealthMax ran three promotions on the Shinez brand in 2022. To identify which to scale in 2024, ROI was calculated for each using the following framework:

Metric	Calculation
Baseline Sales	Average of non-promotional months (via AVERAGE with manual cell selection)
Uplift	Promotional Month Value Sales − Baseline Sales
ROI	(Uplift − Promotion Cost) ÷ Promotion Cost

Value sales for each month were retrieved using VLOOKUP against the Promotion Graph PivotTable. Baseline was locked using absolute cell references ($C$5:$C$7, etc.) to ensure consistent application.

Results:

- Promotion #1: Negative ROI — do not repeat
- Promotion #2 (Buy 2, Get 20% Off): ROI = 12% — highest performing, recommended for 2024
- Promotion #3: Positive ROI but below Promotion #2

3.3 Revenue Forecasting

A 2024 net sales baseline was constructed by forecasting HealthMax's market value sales using Excel's Forecast Sheet (ETS model) from historical data through 2022, then converting forecast market values to net sales using the 2022 ratio:

```
Ratio = Net Sales 2022 / Market Values 2022 Net Sales (Forecast Year) = Forecast
Market Value × Ratio
```

Assumption: The ratio of HealthMax net sales to total market value sales remains stable from 2022 forward — i.e., no change in market share or pricing mix relative to the market is assumed in the base case.

Forecasted 2024 baseline net sales (organic growth only) = $20,808,712.

3.4 Waterfall Chart: Full NRM Impact

All revenue initiatives were compiled into a waterfall chart to communicate the cumulative impact of NRM activities to management:

Component	Value	Type
Estimated Net Sales 2023	$20,480,715	Baseline
Natural Growth	+$327,997	Organic

Organic Shampoo Launch (Herbashine)	+$336,897*	NRM Initiative
50ml Shampoo Launch	+$265,960	NRM Initiative
Estimated Net Sales 2024	$21,411,569	Total

**Approximate; based on Herbashine net sales model from Chapter 2.*

Key finding: NRM-driven initiatives ($602,857) outpace organic/natural growth ($327,997) by approximately 1.8×, validating the strategic value of the NRM framework.

Key Outcomes & Results Summary

KPI / Output	Result	Chapter
HealthMax brands in market	2 (Starbust, Shinez)	Ch. 1
Strongest YoY growth year	2020 (+4% category; +7.7% Shinez)	Ch. 1
Total category MAT (March 2023)	$98,452,906	Ch. 1
HealthMax Net Sales 2022	$20,480,714.90	Ch. 2
Portfolio Gross Margin	68.49%	Ch. 2
Fastest-growing subcategory	Organic (+275.88% since 2018)	Ch. 2
Recommended new product	Herbashine (higher gross profit)	Ch. 2
Best promotion	Promo #2 Buy 2 Get 20% Off (ROI: 12%)	Ch. 3
50ml SKU net sales	$265,960	Ch. 3
2024 NRM-driven incremental sales	$602,857	Ch. 3
Estimated 2024 total net sales	$21,411,569	Ch. 3

Tools & Methodologies

Technique	Application
SUMIFS (multi-criteria)	YTD and MAT calculations across Brand × Region × Year × Month
PivotTables	Market share, YoY growth, subcategory sizing, promotion analysis
Show Values As (% Diff / % of Row)	YoY growth rates and market share derivation
VLOOKUP	Pulling monthly sales data into promotion analysis table
Scatter Plot (Profitability Matrix)	Two-dimensional portfolio prioritization by margin and contribution
Forecast Sheet (ETS)	Time-series extrapolation of market sales to 2024
Waterfall Chart	Visualizing NRM initiative impact on 2024 net sales
Slicers	Dynamic regional filtering on market share charts

Key Assumptions

- 2023 YoY comparisons were excluded due to partial-year data (January–March only)
- MAT is defined as the rolling sum of the most recent 12 calendar months
- The 2022 Net Sales / Market Value ratio is assumed constant for 2023–2024 forecasting
- Organic subcategory is projected to grow 20% from March 2023 MAT levels to full-year 2024
- 50ml SKU launch volume is estimated at 10% of the equivalent 100ml product's 2022 volume
- NRM growth initiatives are assumed to be non-cannibalistic (independent additive contributions)
- Retail price premium for 50ml format is set at 150% of the per-ml price of the 100ml pack